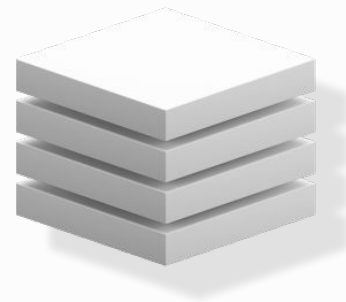




Efficiency Verification of **Multi-Company** **Price**

Minseo Kim, Jihong Min, Dayoon Lee



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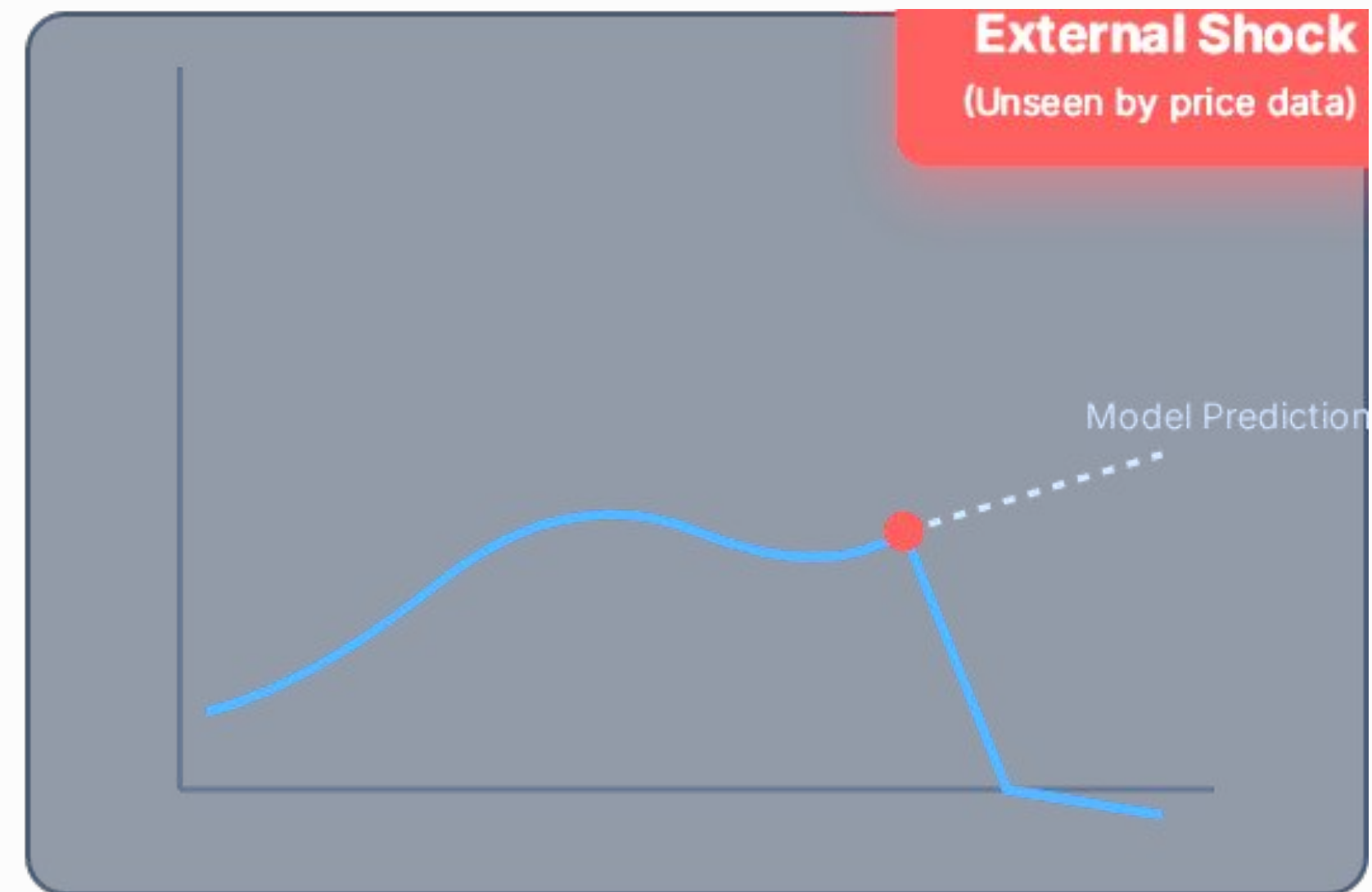
Takeaways

RESEARCH QUESTION

THE BLIND SPOT OF PRICE-ONLY MODELS

Traditional models rely solely
on historical lag data

They cannot see "External Market Conditions"
(News, Macro Events).



RELATED WORK & LIMITATION

To address this, researchers adopted NLP-based methods to directly read the market.

LLM-Augmented Models

Zhou et al. (2025)

Uses LLMs to generate summaries from massive financial texts to feed into Transformers.

Sentiment Analysis

Common Approach

Extracts positive/negative sentiment scores (e.g., FinBERT) from news to adjust price predictions.



CRITICAL LIMITATION: RELY ON EXTENSIVE TEXT SOURCE

Processing massive unstructured text is computationally expensive, suffers from API costs, and has high latency.

Roles

Minseo

Reproduce **NLP-based** method
(LLM / sentiment-based)

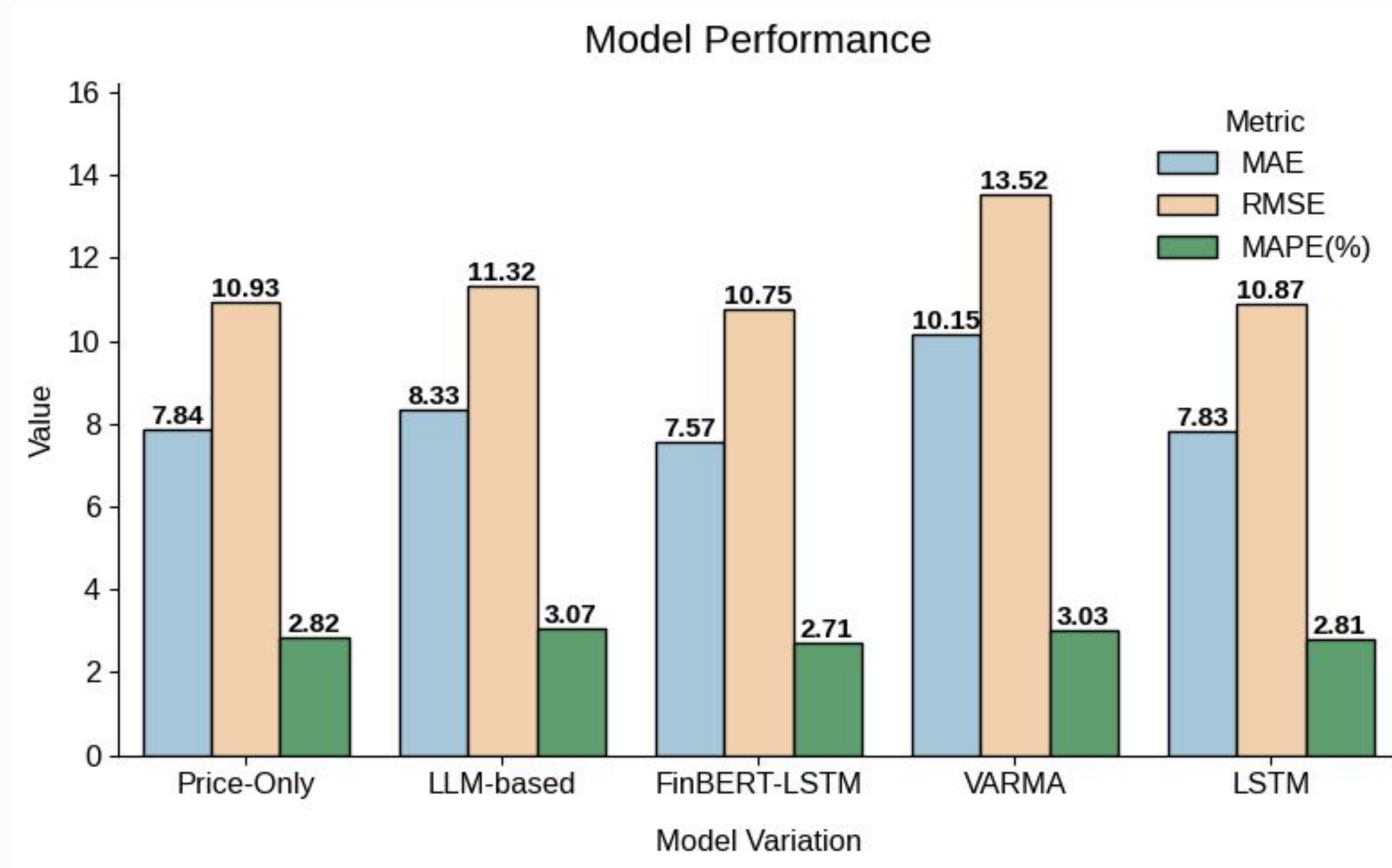
Jihong

Find **Correlated companies** to TESLA /
Apply **VARMA** to multiple companies

Dayoon

Find **Correlated companies** to TESLA /
Apply **LSTM** to multiple companies

Results



FinBERT-LSTM > Multi-Company LSTM > Price-Only > LLM-based > VARMA

Conclusion & Limitations

01

Price-only model underperforms **failing** to capture the **external event**

02

Although **FinBERT-LSTM** method **outperforms** Multi-company based LSTM, it requires **expensive text resources**.

03

Multi-company based method shows **comparable** performance while requiring less resource



Didn't use high-quality news for NLP-based method, gap between NLP-based method and Multi-company based method might be larger

Takeaways

01

Collecting **task-relevant features** can increase performance

02

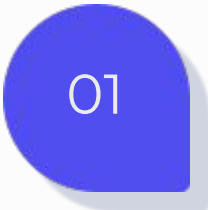
LSTM demonstrates strong **effectiveness**
for time-series forecasting tasks.

STEP-BY-STEP GUIDE

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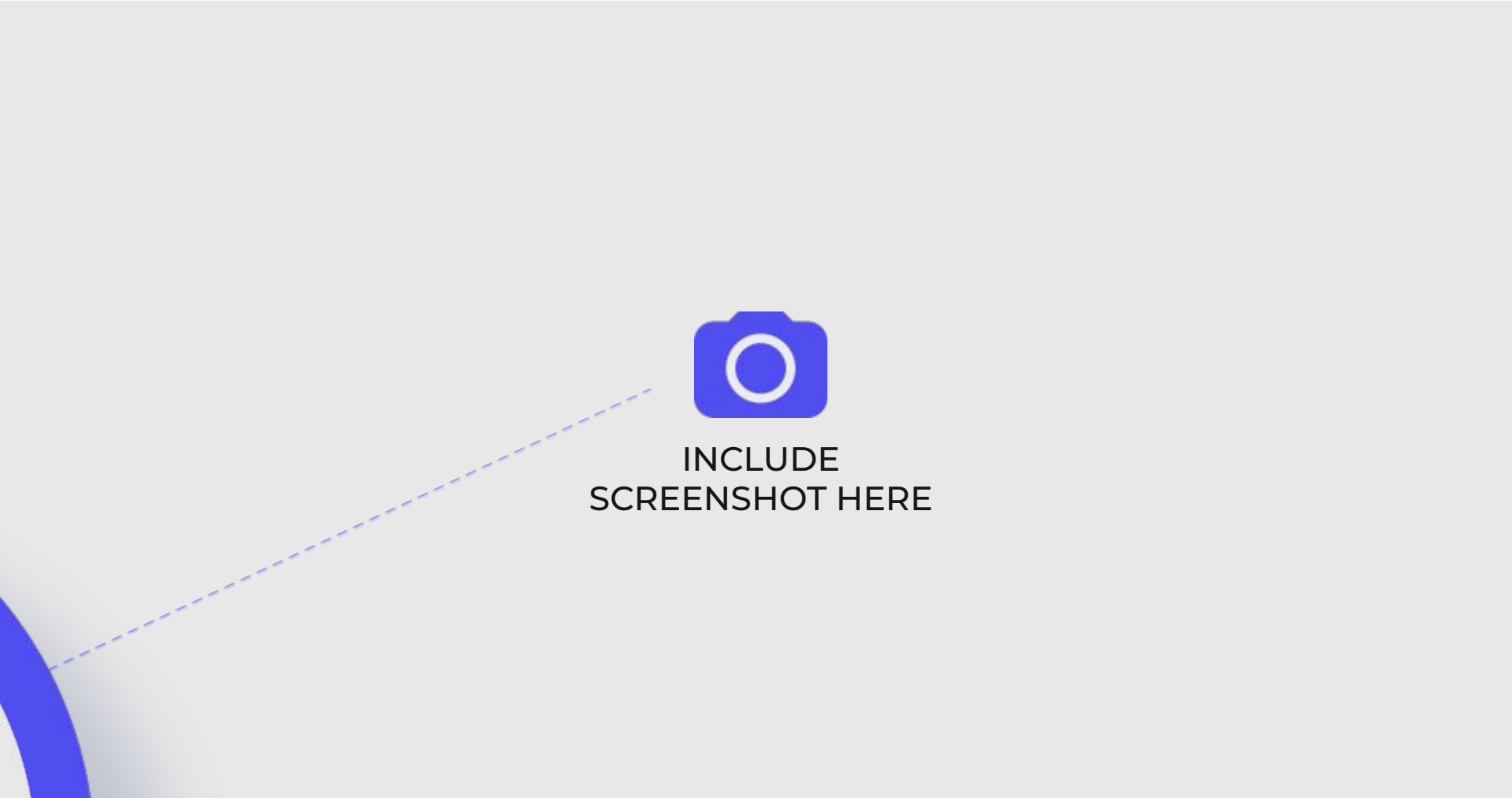
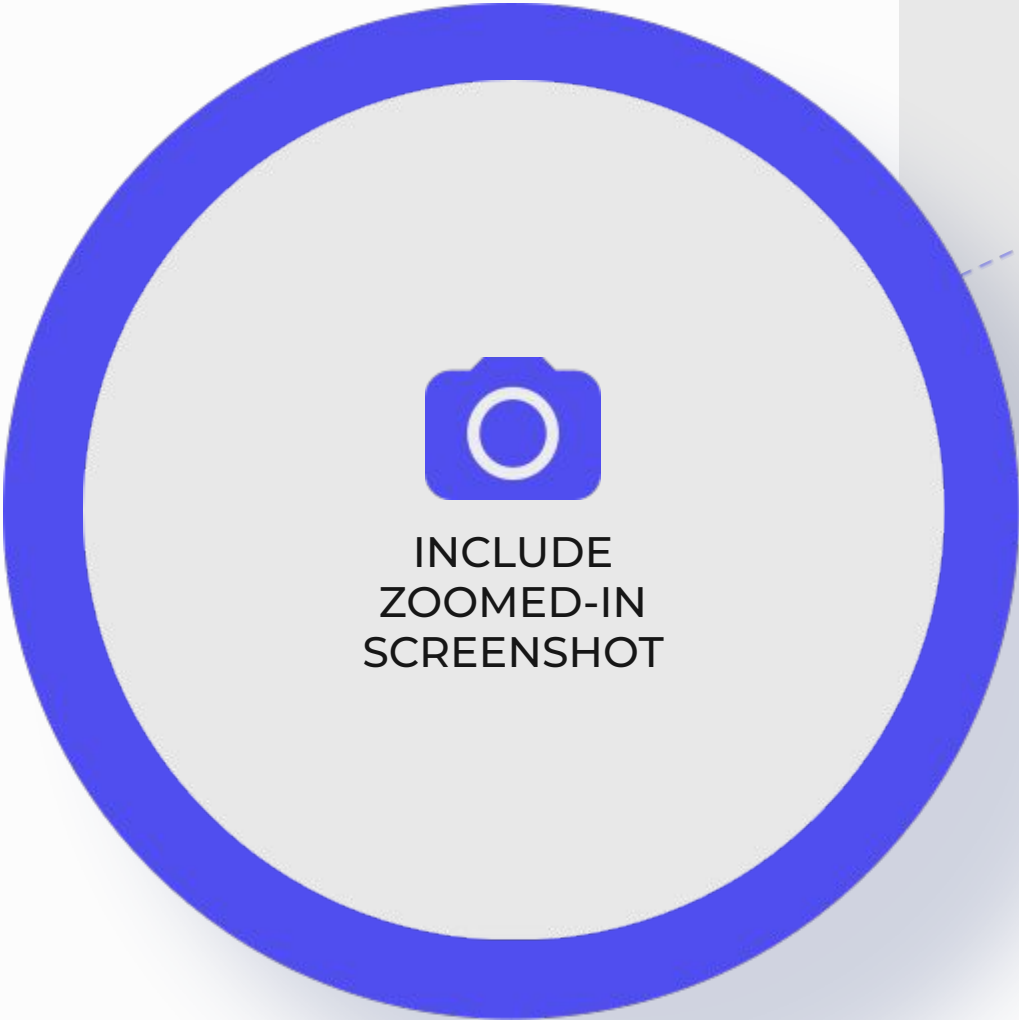
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SUMMARY

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